

Question:

Can I calculate the distance to an object using a camera and a laser?

Hypothesis:

If I shine a laser that is a known distance and angle from a video camera at an object, then I should be able to properly calculate the distance from the camera to the object.

Real world application:

The technique can be used to:

- Scan 3d objects without touching them
- Measure distances without touching an object,:
- Finding distances for self-driving cars

Variables:

Independent: Laser angle, Laser distance from camera

Dependent: distance from camera to object calculated from python program

Control variable: measured distance from camera to object

Research:

1. How to find the camera's frustum angle (field of view)
2. How to code in python
3. How to use OpenCV to obtain video image
4. How to design a model using Fusion 360 and print something with a 3D printer
5. How to find equations of a line
6. How to interpolate
7. How to find the intersection of two line segments
8. How to find the laser dot in an image using OpenCV libraries
9. (slides of how we did the math)
10. How to use the HSV color space to filter out laser dot.

Introduction:

Without using a measuring tape how does one find a certain distance? To calculate the distance to the moon scientists shined a laser at reflective mirrors placed on the moon and calculating the distance based on the speed of light and how long it took for the reflection to reach earth again. However I am applying a different method that can be used in everyday situations where distances are much shorter. With the aid of a program called OpenCV, I use its computer vision to track the location of the laser in the camera's field of view. From this I can mathematically calculate the distance based on the laser's angle as well as distance from the camera using trigonometry. If done correctly, this method can be used in a fast-approaching technology: self-driving cars. Currently, they use a similar method with LiDAR, a device that reflects an invisible laser at objects in order to map out their area as well as distance, instead of calculating the trigonometric distances however, it uses a sensor to detect the laser's reflected pulses. Through this experiment, I aim to find whether or not it is possible to reliably find the laser's distance using a simple camera and laser

Color filtering:

After we capture the video and apply a vertical flip and a gaussian blur (this will help smooth out the laser color so it's easier to detect), we convert the color palette to the HSV color space (hue, saturation, and variance). This allows us to more specifically change these aspects of the color as opposed to RGB which only allows the user to manipulate the redness, greenness or blueness of a color. Then after some experimentation, we adjust these values in order to match the HSV color range of our laser beam. We put this information in a mask that blackens everything not in our specified color range. The program then calls information from our mask window and draws contours over the laser beam. Then it calculates the minimum and maximum x points defined by the contours and draws a rectangle over it. The center of the rectangle is then used as the x coordinate of the found laser position.

Data Analysis:

Judging from the graphs the accuracy is highest at 60 degrees and diminishes at angles further from 60 degrees. We can also see that the accuracy is the highest at the angle's y intercept this is because prior to my measurement I had calibrated the laser in order to match that angle. When the distance is less than the y intercept, the output is said to be further than the actual target. Likewise when the distance is greater than the y intercept, the output is smaller than the actual distance. Because of this, 90 was the most inaccurate since it has no y intercept, being a straight line.

The distance from laser to camera, and the camera angle greatly affects the range that this system can measure.

The further away from the camera, the measurement resolution had gone down.

Conclusion:

This experiment concludes that there can be a lot of issues that come with calculating the distance through pointing a laser and using an off the shelf USB camera. For example, the camera I used was not fixed and was accidentally moved several times. Because of this, I had to calibrate the camera to the correct angle between tests which may have contributed to the inaccuracy. Despite all this, the output stayed in range of 1.5 inches of the actual distance. Another problem I encountered was calibrating the camera. It appeared to be that the lens inside of camera was tilted at a 5 to 10 degree angle! This was due to manufacturing errors. The other problem is the camera could detect many false readings, such as laser reflections off other objects (such as my kitchen table), and other red colors that fell into the lasers HSV color space.

There were also inaccuracies due to my ability to properly measure angles and distances using a standard tape ruler. Having a very slight angle offset results in large distance variations. However, if properly setup, and I had more sophisticated measuring instruments, I believe that this is a viable measurement technique considering all of the other factors. I also believe that with a properly mounted camera and laser, I could remove the errors by creating a lookup table of error corrections.

Also the desired measurement range should be optimized by choosing the best distance and angle for the laser beam for the best measurement resolution.









